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ChatGPT-3.5 and -4.0 and mechanical engineering: Examining performance on the FE mechanical engineering and undergraduate exams

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Abstract

The launch of Generative Pretrained Transformer (ChatGPT) at the end of 2022 generated large interest in possible applications of artificial intelligence (AI) in science, technology, engineering, and mathematics (STEM) education and among STEM professions. As a result many questions surrounding the capabilities of generative AI tools inside and outside of the classroom have been raised and are starting to be explored. This study examines the capabilities of ChatGPT within the discipline of mechanical engineering. It aims to examine the use cases and pitfalls of such a technology in the classroom and professional settings. ChatGPT was presented with a set of questions from junior- and senior-level mechanical engineering exams provided at a large private university, as well as a set of practice questions for the Fundamentals of Engineering (FE) exam in mechanical engineering. The responses of two ChatGPT models, one free to use and one paid subscription, were analyzed. The paper found that the subscription model (GPT-4, May 12, 2023) greatly outperformed the free version (GPT-3.5, May 12, 2023), achieving 76% correct versus 51% correct, but the limitation of text only input on both models makes neither likely to pass the FE exam. The results confirm findings in the literature with regard to types of errors and pitfalls made by ChatGPT. It was found that due to its inconsistency and a tendency to confidently produce incorrect answers, the tool is best suited for users with expert knowledge.

KEYWORDS

ChatGPT, engineering education, fundamentals of engineering exam, mechanical engineering, scientific literacy

1 | INTRODUCTION

The introduction of the electronic calculator revolutionized the jobs of engineers. The tedious and time-consuming task of consulting trigonometric and logarithmic tables was reduced to pressing a few

buttons. In much the same way, the calculator also changed the engineering classroom for students and teachers. The electronic calculator is just one of many disruptive technologies introduced in the 20th and 21st centuries that altered the landscape of the engineering workplace and classroom. Computer-aided design

(CAD) software changed the process of engineering drafting. Personal and high-performance computing changed the timescales needed for complex calculations and simulations. The modern internet remade the platforms used for communication and collaboration. Once the utility of these disruptive technologies is understood by the engineering workforce, their adoption becomes ubiquitous, and as a result, they also become an important element of engineering education. University students around the world are entering classrooms with calculators and laptops in hand, collaborating on learning management systems, and engineering software such as MATLAB, Ansys, and SOLIDWORKS are being directly integrated into the classroom curriculum.

Recent developments in publicly available artificial intelligence (AI) technologies, specifically large language models (LLMs) may be the next disruptive technology that will reshape the engineering profession and engineering education. Generative AI, made possible through LLMs, such as Generative Pretrained Transformer (ChatGPT), has seen tremendous growth in usage recently, and the profession has begun to react. Academic journals are implementing policies around ChatGPT authorship, universities around its usage inside and outside the classroom, companies are determining how to integrate it into workflows, and in some cases, countries are deciding whether to restrict or ban access [16, 43, 45]. Since its release researchers have tested ChatGPT's capabilities in a variety of fields. In this paper, we look to expand on the work of these researchers by testing ChatGPT's ability to answer mechanical engineering questions. Our research intends to address the following questions: How does ChatGPT perform in the field of mechanical engineering, and what use cases may exist for it in the classroom or in professional settings? To this end first, we will share a literature review that provides important background information. Next, we will explain how we determined what questions to ask ChatGPT and how we collected our data. Following the methodology, we will present the results of this study, highlighting the difference between the freely available ChatGPT model and the subscription GPT-4 model (May 12, 2023 versions). Finally, we will conclude the paper with a discussion of the results, the context we believe is important for integrating ChatGPT into engineering education, and future directions of research we plan to pursue.

2 | LITERATURE REVIEW

2.1 | ChatGPT background

Within the first week of its launch on November 30, 2022, ChatGPT, developed by OpenAI [33], reached

1 million users, making it the fastest-growing application to date [7]. ChatGPT is a language model capable of understanding text and producing human-like responses [19]. LLMs like ChatGPT are trained on massive text data sets and aim to predict the next word in a sequence [24], as cited in [19]. The Transformer architecture supports this process by identifying patterns in how words, phrases, and sentences relate to each other and serves as the foundation for ChatGPT's ability to generate text [19, 31]. In contrast to supervised language learning models, OpenAI's use of semi-supervised learning has allowed ChatGPT to leverage large quantities of unlabeled diverse texts, such as Wikipedia articles, in its unsupervised pretraining phase without the need for the additional data processing and labeling required for supervised learning tasks, such as classification [6, 41, 47]. After pretraining, supervised learning is then used to fine-tune the model based on specific natural language understanding tasks [41]. Between the first iteration of GPT in June 2018 and the release of GPT-3 in May 2020, the model's parameters and pretraining data have grown significantly from 117 million parameters and about 5 GB of pretraining data to 175 billion parameters and 45 TB of pretraining data [47]. GPT-4 was released in March 2023 as a large multimodal model that is capable of accepting both image and text inputs; however, image input is not currently available for public use [36]. Alongside the release of GPT-4, OpenAI's release notes for ChatGPT indicate that it is far from a static tool and show that the company releases updates and adds functionality to ChatGPT roughly every 2 weeks [29]. OpenAI also launched a ChatGPT app for iOS devices in May 2023 and has announced plans to incorporate third-party plugins, such as the technical computing Wolfram plugin, with ChatGPT as well as an experimental model that "knows when and how to browse the internet" [35, 38].

2.2 | ChatGPT and exams

OpenAI and researchers from various disciplines have already begun testing ChatGPT's performance on standardized exams and its capacity to solve complex problems. This review will focus on ChatGPT's performance on science, technology, engineering, and mathematics (STEM)-related exams and data sets. While methodologies, the version of ChatGPT used (i.e., GPT-4, GPT-3, ChatGPT, etc.), and the decision to use prompt engineering to improve performance vary among these studies, the results provide valuable insight into ChatGPT's strengths and weaknesses.

Regarding medical exams, Humar et al. [18] found that while ChatGPT did not pass the US Plastic Surgery In-Service Exam, it had “an impressive baseline medical knowledge” but was unable to answer questions that required complex and context-dependent decision making (p. 3). Fijačko et al. [11] similarly found that while ChatGPT was able to provide useful explanations in its answers, it failed to pass the American Heart Association Basic Life Support and Advanced Cardiovascular Life Support exams. Kasai et al. [21] gave both GPT-4 and ChatGPT questions from the Japanese Medical Licensing Exam in Japanese and found that while GPT-4 outperformed ChatGPT and passed the exam, it underperformed compared to medical student averages and selected “prohibited” answers that would harm patients, adding to the notion that ChatGPT and GPT-4 struggle to make complex decisions. Gilson et al. [15] found that while ChatGPT was able to pass the US Medical Licensing Exam, achieving a score equivalent to a third-year medical student, its accuracy was significantly lower for more difficult questions. Fergus et al. [10] further supported that ChatGPT struggles to solve questions that require complex analysis and application of knowledge in a pharmaceutical sciences setting.

In their analysis of ChatGPT's capabilities, Laskar et al. [25] found that ChatGPT performs best in social science and worst in STEM and suggested that its performance in math revealed its tendency to answer questions correctly but with faulty reasoning or with the correct reasoning but the wrong answer. Shakarian et al. [44] also identified ChatGPT's weakness in math, finding that as the complexity of a math word problem increases (e.g., higher numbers of additions and subtractions required), so does ChatGPT's failure rate. This study also found that there was a significant improvement in performance (60%) when ChatGPT was asked to show its work, showing that the use of prompt engineering or modifying prompts can produce clearer answers from ChatGPT [44]. Plevris et al. [39] further emphasized that while ChatGPT was able to provide convincing answers, it did not produce reliable or consistent results on complex mathematical problems or advanced logical tasks and often provided long and detailed incorrect answers without acknowledging any possible errors. Azaria et al. [2] also stated that “ChatGPT demonstrates overconfidence in the mathematical field that surpasses its actual capabilities” (p. 12). Xuan-Quy et al.'s [48] results also found that ChatGPT confidently provided false information and statements on a national high school physics exam in Vietnam and stated that ChatGPT struggled to answer questions that require a deep understanding of physics. Yuan et al.'s [49] analysis found that while ChatGPT was capable of answering

arithmetic questions correctly, in some cases, it often provided estimations that were close to the correct answer, revealing it did not actually calculate the mathematical answer. In their research on how LLM behavior changes over time, Chen et al. [5] found that between the March 2023 and June 2023 versions, GPT-4 performed significantly worse (from 97.6% to 2.4% accuracy) when asked to identify whether a number was prime while GPT-3.5 performed significantly better (from 7.4% to 86.8% accuracy).

Frieder et al.'s [12] broad analysis of ChatGPT's mathematical capabilities further supported the model's limitations, stating that it performed significantly below the average mathematics graduate student, answering questions inconsistently and confidently providing incorrect answers. In their assessment of ChatGPT's ability to solve real-world problems in oil and gas industrial engineering, Ogundare et al. [32] found that ChatGPT faltered when tasked with applying knowledge to novel or unusual scenarios. Giannos and Delardas's [14] assessment of ChatGPT's performance on UK standardized admissions exams also found that ChatGPT underperformed in science and math and when asked questions that require content-specific or prior knowledge of a given topic. Pursnani et al. [40] found that while GPT-4 was able to achieve about 75% accuracy on questions from the Environmental Fundamentals of Engineering (FE) exam, it failed when answering complex multistep calculations. When the correct formula was selected, GPT-4 still arrived at the wrong answer [40]. Koubaa et al.'s [23] analysis of ChatGPT's performance on challenges pulled from the IEEEExtreme computer programming competition emphasized that “ChatGPT still has limitations in understanding and solving complex programming problems that require high-level reasoning and expertise” (p. 7).

In their evaluation of GPT-4 (both with and without visual input), OpenAI [36] highlighted that GPT-4 outperformed its predecessor, GPT-3.5, and achieved high passing scores on STEM standardized exams, such as AP Biology, AP Chemistry, AP Calculus BC, AP Physics 2, and the SAT and GRE math and quantitative sections. While GPT-4 outperformed its predecessor and other LLMs, the model was still prone to several limitations that include: hallucinating (fabricating information), confidently providing wrong answers, lack of knowledge past September 2021, an inability to learn from experience, and “various biases” [34]. Bordt and von Luxburg [3] stressed that while GPT-4 outperformed GPT-3.5 on an undergraduate computer science exam, its ability to pass the exam “[did] not imply that it has any understanding of computer science” (p. 6). Borji [4] corroborated these sentiments, stating that models like

ChatGPT lacked the “capability to reason about the connections between concepts and entities [and] can only generate text based on the patterns they have learned during training” (p. 3).

These studies indicated that while ChatGPT was capable of passing several exams or performing to various degrees of competency on a variety of data sets, it was not a perfect tool. As a language model, ChatGPT did not appear to actually perform mathematical calculations, but instead predicted the answer in the same manner it predicts other text responses [22]. Combining this with ChatGPT's verbose and confident responses, inconsistency, and its tendency to hallucinate results in misleading or wrong answers that appear correct adds to its limitations when answering complex STEM questions. Additionally, ChatGPT struggled with applying theory and knowledge to unique or novel situations, which further supported the notion that it had no real understanding of concepts [12, 32].

2.3 | ChatGPT and engineering education

ChatGPT has already had a disruptive impact on education and scholarly research, with several universities and schools debating how to address potential student use and researchers attempting to list ChatGPT as a coauthor on scholarly research articles [17, 45, 46]. This section will focus on ChatGPT's performance as a “student” in STEM courses and will outline instances where ChatGPT can be used as a supplemental tool to help students and instructors and instances where it may cause more harm than good.

Deshpande and Szefer [8] assessed ChatGPT's performance in an introductory computer engineering course and found that while it performed well on questions related to general concepts, its inability to process or produce images and difficulty with hands-on labs made it incapable of passing the course. Kortemeyer [22] found that ChatGPT was able to narrowly pass an introductory calculus-based physics course but emphasized its inconsistency and frequent numerical errors when performing calculations. While ChatGPT achieved a B- in an introductory functional programming course, Geng et al. [13] emphasized that without prompt engineering, the AI performed significantly worse and highlighted that even for students who opted to use ChatGPT to complete their homework, there was no significant improvement in their homework grades. Elder et al. [9] similarly found that while ChatGPT passed their sophomore-level design course with a 73%

grade, it performed best with follow-up questions and additional explanations for more complex problems.

Increasing dialog among engineering educators, developing best practices and guidelines for generative AI use in the classroom, and establishing an emphasis on training for using these tools can lead to positive applications of ChatGPT in engineering education [20]. ChatGPT may be a useful resource for developing syllabi, creating lesson plans, improving assignments, providing students with personalized feedback, and creating virtual learning environments—with the caveat that its responses can be inconsistent and inaccurate [2, 13, 28]. It may also be possible to use ChatGPT to create tailored problem sets to challenge students at various proficiency levels [2]. ChatGPT can also be a useful test prep resource capable of providing test takers with more meaningful and detailed feedback than traditional test prep materials [11]. As ChatGPT has been shown to be prone to biases and hallucinations, researchers have suggested employing more openness and transparency into the sources of ChatGPT's training data, ChatGPT's design, and how it works can help students and educators understand its limitations and how to effectively use ChatGPT [4, 19, 47].

ChatGPT's negative impacts on engineering education may be more subtle than the concerns educators have on its potential to weaken students' writing skills and its impact on plagiarism. Chegg, a website that provides homework help, and other similar online resources have existed long before ChatGPT's launch and have already served as tools for students to find answers to homework and other problems, dulling the impact of ChatGPT as an outlet for cheating [22]. ChatGPT's current inability to process or generate images also serves as a major limitation to its use in engineering and related subjects (this may change if GPT-4's image processing capabilities are made available through OpenAI's paid ChatGPT Plus service or its free version of ChatGPT) [8, 37]. There are, however, concerns that overreliance on ChatGPT might weaken students' critical thinking and problem-solving skills and may lead to the passive acceptance of ChatGPT's answers as facts [13]. ChatGPT has been inconsistent when providing references and would not be suited for beginners in a given subject as they may be misled into believing ChatGPT's incorrect but overly confident and heavily detailed answers [2, 22, 39, 42]. Language models trained on human text are also imbued with the language humans use with regard to specific topics. ChatGPT, GPT-3, and GPT-4 have been shown to harbor negative attitudes towards math and physics, may exacerbate existing math anxiety in students, and may imply that certain groups are better at math than others, adding to the existing stereotype threat these students may already

experience [1, 27]. Particular care appears necessary when using ChatGPT in the engineering classroom to address its biases and limitations, especially if presented as a tool for beginning learners.

3 | METHODS

The objective of this paper is to determine how successful ChatGPT is as a tool for answering engineering questions to better understand how it and other AI tools may impact engineering education in the future. To this end, we searched for a set of questions that could be representative of one or more fields of engineering, that had unambiguous answers, and that were representative of the expected knowledge of an undergraduate engineering student. The FE exam produced by the National Council of Examiners for Engineering and Surveying [30] “is generally [a] first step in the process of becoming a licensed professional engineer” [30]. The test is 6 h long and consists of 110 questions and is intended for college engineering students nearing graduation or recent graduates. There are versions of the test for a number of different engineering disciplines including but not limited to chemical, mechanical, and environmental engineering. The prominence of the exam also means numerous practice exams and review books have been created, which provide a large set of potential questions. For these reasons, it was determined that using questions from an FE practice exam would be the best way to move forward.

Using the FE exam was not without limitations. First, there are the numerous subject exams to decide between. As an objective of this paper we intended to analyze the responses produced by ChatGPT beyond the simple correct or incorrect to better understand any limitations of the tool that might exist. With this in mind, it was determined that we should start with the mechanical engineering subject exam as one of the authors received their PhD in the field and currently teaches courses at the junior and senior levels on the subject at a large private university. Second, many engineering questions are accompanied by diagrams, figures, tables, and other visuals that need to be viewed and understood to answer the question. At the time that this study took place, neither the free or paid versions of ChatGPT that were studied had the capacity to include an image input. With this in mind we determined that we would limit ourselves to only using questions that could be completely represented through text. Last, the copyright of the practice exam does not allow the text of questions to be redistributed in a publication, such as this article. As a result, we can only share the aggregated data collected

from the FE exam questions and discuss the trends that were seen. To also provide examples of questions and responses we also included additional questions that were developed by one of our authors for use in their Automatic Control Theory and Heat Transfer course exams.

Recognizing these limitations, the authors developed a strategy for inputting questions to ChatGPT, collecting responses, and analyzing the results. The first step was to acquire a set of questions from an FE practice exam. The acquired set consisted of 100 questions with accompanying solutions. The questions were then analyzed to determine which could not be completely represented through text-only input. This dropped the number to 47 questions consisting of multiple-choice (39), fill-in-the-blank (5), matching (1), and all that apply (2) formats. With the set of askable questions determined, each question was typed into a spreadsheet and categorized by both subject and question type. Questions that contained complex mathematical formulas were typed using LaTeX syntax. There are 14 different subjects included in the mechanical engineering FE exam. Table 1 provides these subjects, the number of corresponding questions that could be formatted for text input, the expected number of questions per subject expected on the FE exam, and the percent represented based on the lowest number of expected questions. The correct answer to each question was also included in the spreadsheet.

It is worth noting that efforts were made to determine if visual inputs could be incorporated into the questions. At the time of the study, there was no way to place an image directly into the chat box, but it was found that if a URL associated with an image on the open web was sent to ChatGPT, it could be asked to describe that image. This feature was very inconsistent. At times the descriptions produced by ChatGPT did not match the image at all. In cases when ChatGPT did recognize the general concept of an image, such as an electric circuit diagram, it still regularly misinterpreted the data in the image. For example, when shown an electrical circuit it claimed resistors in series were in parallel and took several additional prompts to correct this error. We also found that ChatGPT was unable to read images from URLs hosted in a Google Drive or Box regardless of the sharing settings. This reinforced the limitation of this study that only questions that could be formulated through text would be used.

Once all the questions were ready for text input, we began the process of collecting data from ChatGPT. Two different accounts were used for inputting the questions to ChatGPT. One account was restricted to the freely available version (GPT-3.5) of ChatGPT, while the second account used GPT-4 via a paid subscription. In both cases

all questions were asked individually, each question was asked as a “new chat.” This allowed us to keep all answers separate and allowed for potential follow-up after analyzing the response. Once a question was asked, the amount of time it took ChatGPT to complete its response was recorded. The full text of ChatGPT’s response was then recorded and it was indicated on the spreadsheet if the response was correct or incorrect. Once all the questions were asked to both models their answers were further analyzed by the authors. This further analysis was done in an effort to better understand what happened when an answer was wrong, and to examine the explanations associated with correct answers. Following this analysis, we generated some follow-up questions for GPT-4 to determine if follow-up would improve its accuracy. The same process was followed for 13 questions developed by the author for their Automatic Control Theory and Heat Transfer Courses.

4 | RESULTS/DISCUSSION

4.1 | Sample questions

As part of the protocol used in this paper, a set of 13 questions used in exams for a 300-level mechanical engineering course in Automatic Control Theory and a 400-level course in Heat Transfer were given to both the freely available version of ChatGPT as well as the subscription-based GPT-4. Overall, GPT-4 answered nine of 13 questions correctly and the freely available version answered eight correctly. This can be further broken down as six questions that both models got correct, two questions that both models got wrong, three questions that GPT-4 got correct while the other model was incorrect, and two questions that the freely available version got correct while GPT-4 was incorrect. In this paper’s Supporting Information [Appendix](#), we provide Table A that contains the text of four sample questions as well as the responses given by both models. The questions and responses are presented exactly as they were given and received. In Supporting Information [Appendix A](#), we have selected one question that both models got correct, one that both models got wrong, one that GPT-4 got correct while the free version was incorrect, and one with the opposite result. Additionally, the last row of Supporting Information Table A provides a free-response question. In this case, GPT-4 provides a correct answer, while the free version does not. This additional question was included in the table to demonstrate the response to questions types other than multiple choice.

A closer examination of the responses provided in Supporting Information Table A demonstrates the positives and negatives offered by ChatGPT with respect to engineering questions. In the case where both provided the correct answer, a significant similarity can be seen throughout both answers, but GPT-4’s answer provides additional detail compared to the freely available GPT-3.5. Though these additional details do not impact the correctness of the response, they do provide more insight into the concept being tested by the question. This indicates that GPT-4 may be a better tool for a student attempting to better understand engineering concepts.

Looking at the sample question that GPT-4 got correct, but the free version did not, the answers again are very similar. Both GPT models start providing the correct equation necessary to complete the calculations and identify the correct information given the in-question prompt. Moreover, both models are able to correctly set up the equation that needs to be solved, but while GPT-4 correctly solves the equation and converts to the proper units that match the multiple-choice option, the free version incorrectly converts from meters to micrometers, instead converting from meters to nanometers and improperly labels its result as micrometers. As a result, the free version provides the incorrect answer. Unit conversion is a common mistake for an undergraduate student, but not the kind of error we anticipated seeing from ChatGPT. Unit errors like this showed up multiple times with regard to both the sample questions and the FE exam questions. It is important to recognize that both GPT models were guilty of this type of error at some point. From the perspective of a student learning mechanical engineering or a professional using ChatGPT in some capacity, this kind of error in a response could be easily overlooked due to how confidently GPT presented its response.

Moving next to the sample question that GPT-4 got incorrect while the free version was correct, there are again several similarities between both answers. It could even be argued that GPT-4 is as correct as the free version, but since it did not select one of the multiple-choice answers it was marked as incorrect—as it would have been had it left the answer blank of a standardized test. The two models give slightly different definitions of the Biot number that is required for this problem. GPT-4 gives a more general equation in terms of a characteristic length and then substitutes that value with the radius. The free version skips the characteristic length step and provides the Biot number equation directly in terms of the radius. Both models correctly calculate a Biot number based on the equation they found and the numbers given

in the problem. Both models also correctly indicate that their calculation satisfies the necessary condition for the Lumped Capacitance model, which is the objective of this question. The reason that neither model produces a value that matches the choices provided is because both used the radius as the characteristic length. In class, students were taught to determine the characteristic length by dividing the system's volume by its surface area. In the case of a sphere, this would result in the radius divided by 3. The method used by the GPT models is a more conservative method and would not be considered wrong if this was a free-response question instead of a multiple choice. After completing the calculation, GPT-4 recognizes its answer does not match any of the choices provided and includes at the end of its response a note that there could be an error in the question. On the other hand, the free model chooses the correct answer, which is the only answer that both has a number that satisfies the lumped capacitance model and says so, but the model offers no explanation for the difference between its calculated value and the one in the selected answer.

In the sample question where both models were incorrect we again see great similarity between the answers. Both models correctly identify the physical equations needed to solve this problem and set up the equations correctly. In fact, each model gives the same equation for finding the coefficient of heat transfer:

$$h = \epsilon \sigma \frac{T_1^4 - T_2^4}{T_1 - T_2} = (0.8)(5.67 \times 10^{-8}) \frac{573.15^4 - 303.15^4}{573.15 - 303.15}.$$

Solving this formula should give the answer, $h = 16.7$, but both models provide incorrect answers. GPT-4 answered with 10.2 and the free version with 1.36. This is a very interesting result as executing arithmetic is the last place a new user might expect to see an error from a software program, but it is consistent with findings in the literature. This is likely a result of GPT being a predictive algorithm and not a calculator. This is important as people may assume that a software like ChatGPT would have no problem executing mathematical expressions and thus may not check its work, which can result in sharing incorrect information.

Last, we take a closer look at the free-response question in the last row of Supporting Information Table A. This question has multiple parts and, on an exam, would have offered students an opportunity to earn partial credit by demonstrating an understanding of the material even if there were mistakes in calculations. Overall, GPT-4 provides a much better answer to this

question. It completes part A correctly, gives reasonable properties for the materials in the problem, correctly calculates the Reynolds numbers and identifies the flow correctly as laminar. GPT-4 also uses the correct Nusselt number correlation for this flow type and ultimately calculates the heat transfer value to within a reasonable rounding error. On the other hand, the free model provides a less accurate set of material properties. It uses these properties to properly calculate the Reynolds number, but it does not identify the flow regime properly. In fact, it states the problem does not specify the flow regime even though the Reynolds number has been determined, which would have cost a student all the points for that section of the question. In the end it also fails in executing the arithmetic of its final expression for heat transfer. This problem demonstrates what was seen throughout all of the questions supplied to ChatGPT: that GPT-4 almost always provided a better answer with more context than the freely available model.

4.2 | FE exam

An objective of this research was to explore the accuracy of the responses provided by the different models of ChatGPT. Our results showed that GPT-4 answered 36 of 47 questions correctly (76.6%) in a total time of 1853 ± 188 s, approximately 30 min. The freely available version of ChatGPT answered 24 of 47 questions correctly (51.1%) in a total time of 1465 ± 188 s, approximately 24 min. The FE exam allows 6 h for completion and even accounting for the fact that we asked less than half the total questions on the exam it is clear that both models are capable of meeting the allowed time frame. It is not possible to see if either model would pass the exam as the requirements for passing are not available, but it is clear that the accuracy of the models is substantially different, with GPT-4 clearly outperforming the freely available version.

To look deeper into the results we also analyzed the accuracy of responses by subject and by question type as seen in Tables 2–5. It is important to recognize the relationship between Table 1 and Tables 2 and 3 that focus on answers by question subject. Table 1 demonstrates that a number of subjects are underrepresented in this study. This is primarily due to the fact that some subjects are more prone to visual questions that require charts or diagrams and, as a result, could not be transformed into the text-based input required of this study. Six subject areas have less than 50% of their lowest expected number of questions represented: Statics, Dynamics, Mechanics of Materials, Fluid Mechanics, Heat Transfer, and Mechanical Design and Analysis.

TABLE 1 Question subjects for the mechanical engineering FE exam and corresponding number of questions available and asked.

Analysis by question subject	Questions with text only input	Expected number on FE exam	% Represented
Mathematics	4	6–9	67
Probability and Statistics	4	4–6	100
Ethics and Professional Practice	4	4–6	100
Engineering Economics	3	4–6	75
Electricity and Magnetism	3	5–8	60
Statics	0	9–14	0
Dynamics, Kinematics, and Vibrations	2	10–15	20
Mechanics of Materials	1	9–14	11
Material Properties and Processing	6	7–11	86
Fluid Mechanics	4	10–15	40
Thermodynamics	6	10–15	60
Heat Transfer	2	7–11	29
Measurements, Instrumentation, and Controls	4	5–8	80
Mechanical Design and Analysis	4	10–15	40

Abbreviation: FE, Fundamentals of Engineering.

TABLE 2 Analysis of accuracy based on question subject for GPT-4.

Analysis by question subject: GPT-4	Total questions per subject	Right by subject	Percentage
Mathematics	4	4	100
Probability and Statistics	4	4	100
Ethics and Professional Practice	4	4	100
Engineering Economics	3	2	66.67
Electricity and Magnetism	3	3	100
Statics	0	0	Undefined
Dynamics, Kinematics, and Vibrations	2	2	100
Mechanics of Materials	1	0	0
Material Properties and Processing	6	3	50
Fluid Mechanics	4	3	75
Thermodynamics	6	3	50
Heat Transfer	2	2	100
Measurements, Instrumentation, and Controls	4	3	75
Mechanical Design and Analysis	4	3	75

Abbreviation: GPT, Generative Pretrained Transformer.

Three subjects have between 50% and 74%: Mathematics, Electricity and Magnetism, and Thermodynamics. The remaining five subjects are represented at 75% or greater: Engineering Economics, Probability and Statistics, Ethics and Professional Practice, Material Properties

and Processing, and Measurement, Instrumentation, and Control.

Comparing Tables 2 and 3, it is seen that GPT-4 performed as well as or better than the free version in nearly every subject. The only exception was Mechanics

TABLE 3 Analysis of accuracy based on question subject for free version of ChatGPT.

Analysis by question subject: GPT-4	Total questions per subject	Right by subject	Percentage
Mathematics	4	2	50
Probability and Statistics	4	3	75
Ethics and Professional Practice	4	2	50
Engineering Economics	3	0	0
Electricity and Magnetism	3	2	66.7
Statics	0	0	Undefined
Dynamics, Kinematics, and Vibrations	2	0	0
Mechanics of Materials	1	1	100
Material Properties and Processing	6	3	50
Fluid Mechanics	4	2	50
Thermodynamics	6	3	50
Heat Transfer	2	1	50
Measurements, Instrumentation, and Controls	4	3	75
Mechanical Design and Analysis	4	2	50

Abbreviation: GPT, Generative Pretrained Transformer.

of Materials, but as this subject is only represented by a single question in the data set, it would be inappropriate to draw larger conclusions from it. Additionally, a closer examination of the response shows that both versions of ChatGPT calculated an incorrect value for this answer, but the free version chose the correct multiple-choice option stating:

None of the answer choices provided match this value exactly. However, the closest option is:

C. 1,320

Please note that the given answer choices might contain typographical errors or approximations, as the calculated value does not precisely match any of them.

More interestingly, in this case, C was not actually the closest answer to the value ChatGPT produced (2646.75) in the multiple-choice options, but based on our methods, since this was the stated answer, it was marked as correct.

Overall, both versions of ChatGPT demonstrated that they are capable of answering engineering questions. GPT-4 provided accurate answers more often. GPT-4 had its worst performance, outside of the single Mechanics of Materials question, in the Thermodynamics and Material

TABLE 4 Analysis of accuracy based on question type for GPT-4.

Analysis by question type: GPT-4	Total questions per type	Right by category	Percentage
Multiple choice	39	30	76.9
Fill in the blank	5	3	60.0
Matching	1	1	100
Ordering	0	0	Undefined
All that apply	2	2	100

Abbreviation: GPT, Generative Pretrained Transformer.

Properties and Processing. Both these sections were well represented with six questions, but in both subjects, GPT-4 only answered 3 correctly. For many of the wrong answers in these subjects, GPT-4 provided the correct methodology but required a material property or access to a chart or table that would have been included in the FE exam supplemental materials for an exam taker but was not available in the context of this study. In most cases GPT-4 stated that it would need more information to provide the calculations.

Tables 4 and 5 demonstrate that both models are capable of answering multiple-choice, fill-in-the-blank, matching, and all that apply questions. Both models struggled the most with fill-in-the-blank questions

TABLE 5 Analysis of accuracy based on question type for free version of ChatGPT.

Analysis by question type: Free version	Total questions per type	Right by category	Percentage
Multiple choice	39	22	56.4
Fill in the blank	5	1	20.0
Matching	1	1	100
Ordering	0	0	Undefined
All that apply	2	0	0

Abbreviation: GPT, Generative Pretrained Transformer.

(however, our data set is small). This reflects that when calculations were off, both models would sometimes still select a multiple-choice answer that was closest to their calculation, but could not do this for fill-in-the-blank type questions.

4.3 | ChatGPT, assessment, and the classroom

An important question that our research can also help address is how instructors can adapt their classrooms and assessments with regard to student use of ChatGPT and other generative AI technologies. Much like a solution manual or online resources such as Chegg or Stack-Exchange, ChatGPT can help students as they learn and seek to understand novel and challenging concepts or provide insight students can benefit from as they approach solving specific types of problems. However, like Chegg and StackExchange, ChatGPT can also hinder that process and introduce opportunities for cheating on assessments designed to measure students' learning. Similarly, ChatGPT and information on Chegg and StackExchange can also lead students to misunderstand or incorrectly apply concepts if these resources are not critically approached and examined with regard to their limitations and strengths. On the basis of the data compiled in this paper, as well as our experience with ChatGPT and classroom instruction, we propose the following recommendations and insights with regard to classroom instruction and assessment.

Generative AI is an evolving tool. Our data were collected using the May 12, 2023 version of ChatGPT-3.5 and -4.0, and our recommendations are based on those two versions. We fully expect to see the capabilities of ChatGPT and other generative AI tools continue to grow, which may impact the efficacy and relevance of our recommendations. For this reason, our first recommendation is for all

instructors to carefully reflect on their course objectives and to consider if ChatGPT or other generative AI tools should or could be a part of the learning process. As an example, a calculus course may not allow the use of advanced calculators during an exam if the instructor wants to assess the students' ability to perform integrals. On the other hand, a course on heat transfer may allow such a calculator to be used during an exam because the instructor is concerned with the students' ability to apply the integral within the larger context of a heat transfer problem and is less concerned with the students' ability to perform the integration. If AI tools are being used in the engineering workforce, like calculators, it is not by default inappropriate for instructors to allow them to be used for assignments. However, instructors will benefit from critically examining how ChatGPT or other AI tools will interact with their learning objectives, assessments, and pedagogical approaches. Once an instructor is able to determine how ChatGPT or other AI tools' strengths and weaknesses can positively and/or negatively impact their courses and pedagogical styles, we suggest that instructors craft a clear policy on the usage (or restriction) of these tools in their syllabi and explain it at the start of the semester.

While a policy can help clarify expectations around ChatGPT and AI tool usage, instructors may still be concerned with how to enforce these policies to prevent students from using ChatGPT or another AI tool to cheat or bypass the learning process. The remainder of this section will focus on recommendations to address concerns around cheating based on five common engineering assessments: technical problems (on homework or exams); lab reports; research papers; design projects; and coding assignments. These recommendations are summarized in Table 6 at the end of this section.

Technical questions, such as those given as a part of an exam or problem set, are most similar to the FE exam questions used to evaluate ChatGPT's capabilities earlier in this paper. While ChatGPT can successfully answer both multiple-choice and open-ended questions, ChatGPT is also prone to producing hallucinations and misinformed answers, confidently presenting mathematical solutions with significant errors, and lacks the ability to interpret images or diagrams. The best course of action that can be recommended at this time is to utilize visuals in both the questions and answers when possible. At the time of this research, visual inputs were not possible, and though this may change with time, it will add a layer of difficulty that can discourage cheating. Additionally, performing assessments in person without the use of internet-capable devices would make it very difficult for students to use ChatGPT or other AI tools.

Lab reports, research papers, and design projects all contain written sections that can align with ChatGPT's

TABLE 6 Summary of assessment types and recommendations for instructors.

Assessment type	ChatGPT Strengths	Recommendations
Technical problems	Can provide direct answers and produce explanations for solving	When possible, use images or videos and conduct assessments in class
Lab reports	Can assist in writing, explaining methodology, and experimental design	Consider asking for raw data, reflective journals, ask for technical images, and follow-up in-class questions
Research papers	Can assist in writing, finding sources, or summarizing sources	Consider reflective journaling, requiring annotated bibliographies, and follow-up in-class questions
Design projects	Can assist in writing, ideating, and development	Consider reflective journaling, requiring technical drawings, and follow-up in-class questions
Coding assignments	Can assist in producing code	Consider reflective journaling, requiring code annotations, and follow-up in-class questions

ability to produce competent written responses. Although the generation of written reports, papers, and projects was not part of our data collection, ChatGPT's written responses to the questions we did ask demonstrate its ability to produce clear and coherent answers to engineering questions. ChatGPT could, therefore, serve engineering students who seek to avoid writing even if assignments are associated with complex technical concepts and topics related to engineering. ChatGPT can also function as a useful tool to aid students as they formulate research questions or look for background information on a topic, though in our experience, we have seen it produce incorrect citations when asked for sources of information, which can create difficulty for students looking for references. With these considerations in mind, STEM instructors may benefit from learning about the approaches humanities, social sciences, and other writing-intensive disciplines are taking to address ChatGPT's usage in the writing process. Additionally, STEM instructors may also benefit from collaborating with librarians to incorporate concepts of information literacy, such as critically examining a given source of information and finding reliable sources of information, into their courses. While ChatGPT may serve students as they design and explain experimental setups in a lab report, it may also provide a meaningful role in helping students ideate and iterate upon a design while also offering potentially useful suggestions on how to test or improve a given design. It can be exceedingly difficult to determine whether ChatGPT or a related tool was used to generate written responses in lab reports, research papers, and design projects. While written assignments can be "checked" using antiplagiarism tools such as Turnitin, these tools can produce false positives and false negatives. Rather than relying solely on methods that may potentially falsely penalize or absolve students, we recommend the following suggestions. (1) Require students to include technical

images, such as CAD drawings or experimental schematics, with their written assignments when possible. (2) Require students to submit raw data, CAD files, or annotated bibliographies with their reports. (3) Pair lab reports, research papers, and design project assessments with journaling assignments where students reflect on the steps taken to complete the assignment. (4) Incorporate follow-up questions related to these assessments into in-class exams. (5) Encourage students to critically comment or reflect on how they collected information for a research paper or design project and lab report, if applicable.

While the data we collected within this paper does not attempt to quantify ChatGPT's capabilities with regard to coding, related research has explored this question [8]. Our recommendations for instructors concerned with coding assignments are similar to those previously mentioned. Instructors should consider incorporating follow-up questions on coding assignments during in-class exams. Instructors should also consider requiring that all submitted code is annotated and perhaps that students also provide a reflective journal about how they developed the code and how they approached their debugging process.

Nearly every area of higher education has been impacted by the launch and use of ChatGPT, forcing educators and instructors across disciplines to grapple with the impact of ChatGPT on assessments, writing, and student learning overall. While ChatGPT can function as a useful and even beneficial tool for engineering students seeking individualized support as well as for STEM instructors as they develop lesson plans and instructional materials, approaching ChatGPT critically and understanding its limitations is important [13, 28]. Last, while outside the scope of this paper, further research and discussion that examines the use of various pedagogical approaches and the application of critical frameworks in STEM education as they relate to concerns surrounding student use and reliance on ChatGPT may also provide

valuable insights to help adapt STEM education to the age of generative AI.

5 | CONCLUSION

This paper seeks to expand upon literature that has attempted to analyze ChatGPT's capabilities in various disciplines. To that end, we have focused on putting two models of ChatGPT, the freely available version (GPT-3.5) and the subscription-based GPT-4, through the mechanical engineering FE exam. The NCEES does not provide the information needed to determine what it takes to pass the FE exam, but due to ChatGPT's limitation of text-only inputs and the correct response rates seen in our data, we feel it is unlikely that either model would pass the FE mechanical engineering exam. However, it is clear from our data that GPT-4 greatly outperforms its free counterpart. Additionally, as our methodology did not utilize prompt engineering, our approach may mimic how an undergraduate student might use ChatGPT to assist to answer engineering questions, providing information regarding both the potential use cases, as well as pitfalls, of ChatGPT from both student and instructor perspectives.

We found that, compared to its free counterpart, GPT-4 consistently provided more detail in its answers and was 50% more likely to be accurate based on our small data set of engineering questions. We saw that GPT-4 performed well in nearly every topic it was given and could handle a variety of question types. Additionally, GPT-4 was more likely than the free version to avoid making invalid assumptions and attempting to solve problems without all the necessary information. However, both models were capable of providing detailed misinformation. We can confirm what was seen in the literature, where both models made errors in arithmetic and units but often presented those errors with confidence as correct answers. This makes it clear that anyone using ChatGPT to solve engineering problems must be willing to double-check its work. While users can ask follow-up questions to clarify errors in ChatGPT's responses, our work and related research utilizing prompt engineering shows that following up or rewording questions does not necessarily lead to a correct response. At the time of writing this article, ChatGPT's inability to accept visual inputs presents another significant limitation to its utility in handling engineering questions with diagrams or other images. From an instructor's perspective, this may be treated as an advantage, allowing the use of plots, charts, and other visual schematics in homework or exam questions to prevent students from using ChatGPT. These findings support existing literature addressing the use of ChatGPT in other STEM disciplines.

On the basis of our research and the conclusions drawn from the literature we consulted for our review, we have identified possible applications of ChatGPT in the engineering classroom. As an LLM, ChatGPT's ability to pull information from a variety of sources and identify patterns may make it a useful assistant when developing lesson plans and assignments. Educators, as experts capable of verifying the accuracy of resources in their given domains, can use ChatGPT to develop materials for their courses and may be able to use the tool to develop differentiated learning materials for students of varying skill levels. Using ChatGPT to facilitate a flipped classroom approach may also present an opportunity to embrace the technology rather than discourage its use. Engineering faculty may also be able to leverage the tool's inconsistency and inaccuracy to spark discussion among students and sharpen their critical thinking and problem-solving skills [26]. ChatGPT's ability to provide explanations and its broad training data can also make it a useful tool for students seeking background information on general topics and definitions for specific terms with the caveat that students should corroborate their answers with other sources. This may add to students' learning experiences by offering them an outlet to ask questions and receive personalized feedback without the fear (imagined or real) of judgment associated with asking questions in class or during office hours. Last, as Meneske [28] suggests, ChatGPT may also be used to create virtual learning environments and simulations of real-life engineering situations that can help students visualize and apply engineering concepts in novel ways.

We have also identified a possible approach educators can take to mitigate negative impacts associated with ChatGPT's use, such as cheating. As shown in our literature review and in ChatGPT and GPT-4's difficulty in answering multistep open-ended questions from the FE exam and junior- and senior-level course exams, ChatGPT does not perform well when tasked with applying knowledge to unique and novel situations, producing complex analyses that require the application and interpretation of knowledge, and answering context-dependent questions. We have also provided recommendations for how instructors can design assessments that can remain effective in the time of generative AI. Rather than employing punitive approaches to prevent students from using ChatGPT, educators can use the disruptive impact of this tool to reimagine their assignments and create questions and assessments that are difficult for LLMs to process [10]. Encouraging students to be transparent in how they answer questions and focusing on the process of problem solving rather than arriving at a specific numeric answer may also be a useful approach in building trust (i.e., rather than assuming malintent) and

increasing students' understanding of how to develop strong problem-solving and critical thinking skills.

In professional settings, ChatGPT may be a useful assistant for professionals seeking to refresh their knowledge of specific topics and as a supplement to searches for information. However, caution should be exercised when considering ChatGPT as a mathematical tool due to its tendency to confidently produce incorrect answers and its functionality as an LLM that predicts responses rather than performing calculations. Therefore, professionals who intend to use ChatGPT in the workplace may want to avoid using the tool to make automated decisions and include a verification step in any workflow that utilizes the ChatGPT. In both professional and educational contexts, greater transparency around the design, training data, and development of ChatGPT would aid users in understanding the tools' biases and limitations, allowing for more effective and informed use.

Though a big question still left unanswered is how critical students and researchers are of the information they receive from ChatGPT and similar programs. To address this final question, the authors have started a second research study focused on examining how engineering students engage with ChatGPT. The study will be conducted in two parts. The first will be a survey used to gather information on how students are engaging with ChatGPT. The second part will consist of observing students' use of ChatGPT to answer preset questions and then interview those students about their choices and confidence in the results.

While ChatGPT has the potential to be a capable tool for users in many disciplines, including engineering, it is clear that due to its inconsistency and tendency to confidently produce incorrect answers, it is best suited for users with expert knowledge. The introduction of the calculator raised concerns among math educators about whether or not students would be able to learn the mechanisms behind their calculations if a calculator was permitted. Similar concerns were raised with the advent of the PC, increased access to the internet, and, now, with access to ChatGPT. One of the key distinctions between these tools is their reliability. Calculators do not commonly produce incorrect calculations and educators have found ways to incorporate their use in the classroom without compromising students' ability to solve problems. Access to the internet was seen as another disruptive tool and, while the reliability of information on the internet is a constant cause for caution, the use of PCs and the internet has become a commonplace tool in the classroom. ChatGPT may present the next disruptive technology in science and engineering, but it may be premature to claim the

current models as such. Much like the calculator and the PC before it, ChatGPT may be poised as the next opportunity for educators to reevaluate their methods and can make the case for increased critical thinking and information literacy skills in all disciplines. If ChatGPT is here to stay, further exploration and examination of how ChatGPT can or should be incorporated into STEM education and professions is vital.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data related to the FE Exam cannot be shared due to copyright. Other data can be requested from the authors.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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